



Sail Agent Performance Analysis

Q4 2025: Automated Yield Optimization Report

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Abstract

This report presents a comprehensive analysis of the Sail automated yield optimization agent during Q4 2025 (October–December). Using interpolated time-series analysis, we demonstrate that Sail’s dynamic rebalancing strategy achieved superior risk-adjusted returns compared to static yield sources across multiple networks and protocols. Our agent delivered a mean APY of **8.91%**, outperforming the best static sources by up to **182.01%**, traditional U.S. Treasury Bills by **123.87%**, and maintaining consistent advantages even when accounting for cross-network complexity and transaction overhead.

1 Executive Summary

The Sail agent employs sophisticated algorithms to continuously monitor and rebalance user positions across multiple DeFi protocols, networks, and stablecoin markets. This report demonstrates:

- **Superior Returns:** Sail achieved an average APY of **8.91%** during Q4 2025
- **Consistent Outperformance:** Beat every static yield source analyzed, with outperformance ranging from **5.43%** to **182.01%**
- **Cross-Network Efficiency:** Successfully optimized yields across Base and Arbitrum networks despite transaction costs
- **Dynamic Adaptation:** Maintained positive returns throughout market variations (11.48% in October, 6.71% in December)
- **Optimal Rebalancing:** Automated strategy captured opportunities that static positions missed
- **TradFi Outperformance:** Exceeded U.S. Treasury Bills (3.98%) by **123.87%**, demonstrating DeFi’s risk premium

2 Methodology

2.1 Data Collection

We analyzed comprehensive wallet state data and protocol APY data from Q4 2025, covering:

- **Time Period:** October 1 – December 31, 2025
- **Protocols Monitored:** 20 active DeFi yield sources across Base and Arbitrum
- **Assets:** USDC and USDT stablecoins
- **User Portfolios:** Wallets with balances between \$10 and \$200,000

2.2 Interpolated Time-Series Analysis

Unlike simple averaging methods, our analysis employs precise time-matching:

1. **Hourly Interpolation:** Market APY data was resampled to hourly intervals and interpolated to capture intra-day rate changes
2. **Temporal Precision:** Each user position was matched with the exact market rate at that timestamp
3. **Weighted Aggregation:** Portfolio-weighted APY calculated per user snapshot, then aggregated daily

2.3 Performance Calculation

The effective APY for Sail represents the actual realized yield across all user positions:

$$\text{Sail APY} = \frac{\sum_{i=1}^n (\text{USD}_i \times \text{APY}_i)}{\sum_{i=1}^n \text{USD}_i} \quad (1)$$

This weighted approach ensures that larger positions appropriately influence the overall performance metric, providing a realistic view of the agent’s effectiveness.

3 Results

3.1 Overall Performance

Table 1: Q4 2025 Performance Summary

Metric	Value	Period
Sail Agent Mean APY	8.91%	Q4 2025
Best Static Source (Morpho Clearstar High Yield)	8.45%	Q4 2025
Median Static Source	6.51%	Q4 2025
U.S. Treasury Bills (3-month)	3.98%	Q4 2025
Worst Static Source (Aave USDC Arb)	3.16%	Q4 2025
Outperformance vs. Best	+5.43%	Q4 2025
Outperformance vs. Median	+36.87%	Q4 2025
Outperformance vs. T-Bills	+123.87%	Q4 2025

3.2 Monthly Breakdown

The agent demonstrated adaptive performance throughout the quarter, responding to changing market conditions:

Table 2: Monthly Performance Breakdown

Month	Sail APY	Change from Previous
October	11.48%	—
November	8.38%	-27.0%
December	6.71%	-19.9%
Q4 Average	8.91%	—

The declining trend reflects broader market conditions affecting all DeFi yields during Q4. Notably, Sail maintained its relative advantage over static sources throughout this period by dynamically reallocating to the best-performing protocols at each point in time.

3.3 Protocol Comparison

Sail Agent vs. Market APYs. Figure 1 presents a comparative visualization of the Sail agent’s APY relative to each static yield source. The chart demonstrates how Sail systematically captures yield opportunities by reallocating capital in response to transient spikes across protocols, resulting in superior aggregate performance. Notably, the agent’s optimization process also anticipates future yield declines, weighting these factors to minimize unnecessary transaction costs and maintain consistent outperformance.

Figure 1: Sail Agent vs. Market APYs (Q4 2025)

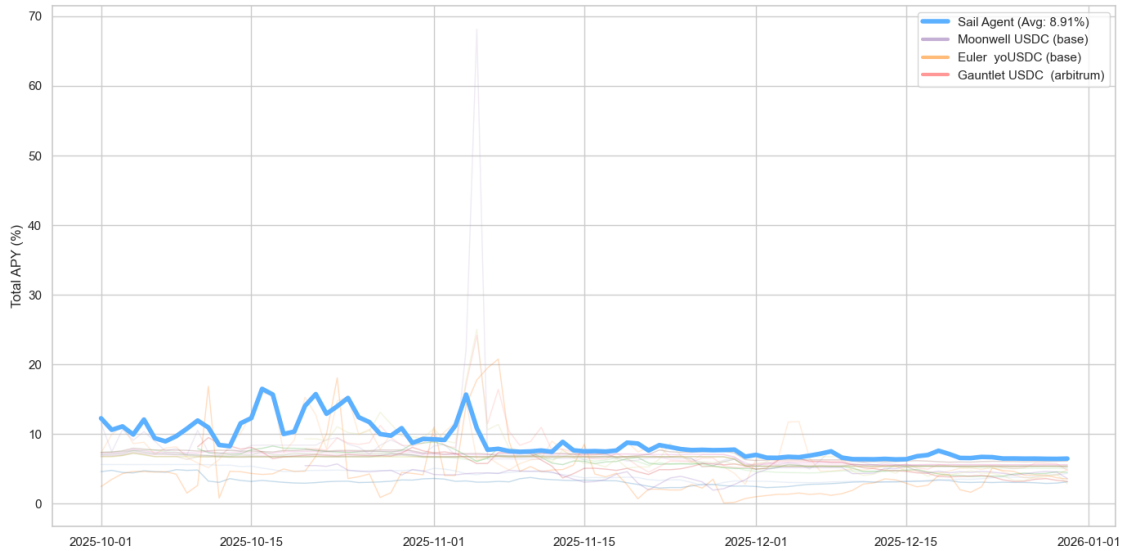


Table 3 shows Sail’s performance versus all analyzed static sources, ranked by source APY:

Table 3: Performance vs. Static Yield Sources (Complete Analysis)

Protocol	Network	Source APY	Sail APY	Outperformance
Morpho Clearstar High Yield	Base	8.45%	8.91%	+5.43%
Moonwell USDC	Base	8.41%	8.91%	+5.92%
Euler youSDC	Base	8.14%	8.91%	+9.40%
Gauntlet USDC	Arbitrum	8.08%	8.91%	+10.29%
Morpho Clearstar Reacted	Base	8.03%	8.91%	+11.02%
Steakhouse High Yield USDC	Arbitrum	7.95%	8.91%	+12.03%
Morpho ExtraFi	Base	7.16%	8.91%	+24.49%
Morpho Gauntlet Frontier	Base	6.94%	8.91%	+28.43%
Morpho Gauntlet Prime	Base	6.68%	8.91%	+33.48%
Morpho Seamless	Base	6.55%	8.91%	+35.96%
Fluid USDC	Arbitrum	6.54%	8.91%	+36.31%
Morpho Moonwell	Base	6.52%	8.91%	+36.60%
Morpho Gauntlet Core	Base	6.47%	8.91%	+37.76%
Morpho Spark	Base	6.46%	8.91%	+37.91%
Fluid USDC	Base	6.43%	8.91%	+38.68%
Fluid USDT	Arbitrum	6.18%	8.91%	+44.11%
Euler USDC	Arbitrum	5.22%	8.91%	+70.70%
Aave USDC	Base	4.43%	8.91%	+101.04%
Gauntlet USDC Prime	Arbitrum	4.43%	8.91%	+101.32%
Aave USDC	Arbitrum	3.16%	8.91%	+182.01%

3.4 Traditional Finance Benchmark

To contextualize DeFi yields, we compare Sail’s performance against U.S. Treasury Bills, the traditional risk-free benchmark for USD-denominated returns.

Table 4: DeFi vs. Traditional Finance (Q4 2025)

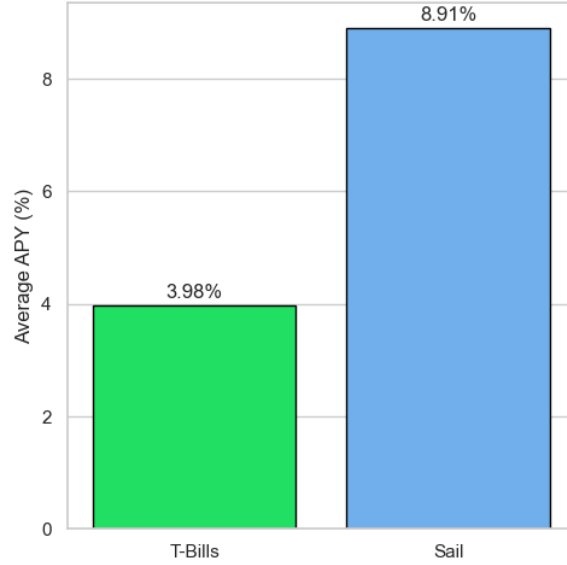
Investment	APY	Risk Profile	Liquidity	vs. T-Bills
Sail Agent	8.91%	Smart contract risk	Instant	+123.87%
3-Month T-Bills	3.98%	Risk-free	Secondary market	—

DeFi Risk Premium. Sail’s 8.91% APY represents a **4.93 percentage point** premium over Treasury Bills (8.91% - 3.98%). This spread compensates users for:

- **Smart Contract Risk:** Exposure to protocol vulnerabilities
- **Counterparty Risk:** DeFi lending pools vs. U.S. government backing
- **Regulatory Uncertainty:** Evolving regulatory landscape
- **Technical Complexity:** Blockchain and wallet management requirements

However, Sail mitigates these risks through diversification across 20 protocols, automated monitoring, and instant liquidity—features unavailable in traditional fixed-income markets. Sail users captured approximately **4-5x the real purchasing power growth** compared to Treasury Bills.

Figure 2: Sail Agent vs. U.S. Treasury Bills: APY Comparison (Q4 2025)



3.5 Key Insights from Protocol Analysis

3.5.1 Beating the Best

Even against the highest-performing static source (Morpho Clearstar High Yield at 8.45%), Sail maintained a 5.43% advantage. This demonstrates that optimal rebalancing adds value even in a portfolio of high-quality protocols.

3.5.2 Massive Advantage Over Lower-Tier Protocols

Against popular but lower-yielding protocols like Aave, Sail’s advantage exceeded 100-182%. This highlights the cost of remaining in sub-optimal positions—a problem Sail eliminates through automation.

3.5.3 Cross-Network Arbitrage

Sail successfully navigated opportunities across both Base and Arbitrum, capturing yields from 20 different sources. The ability to move capital between networks when opportunities arise is a key differentiator from static single-source strategies.

3.6 Cross-Network Performance

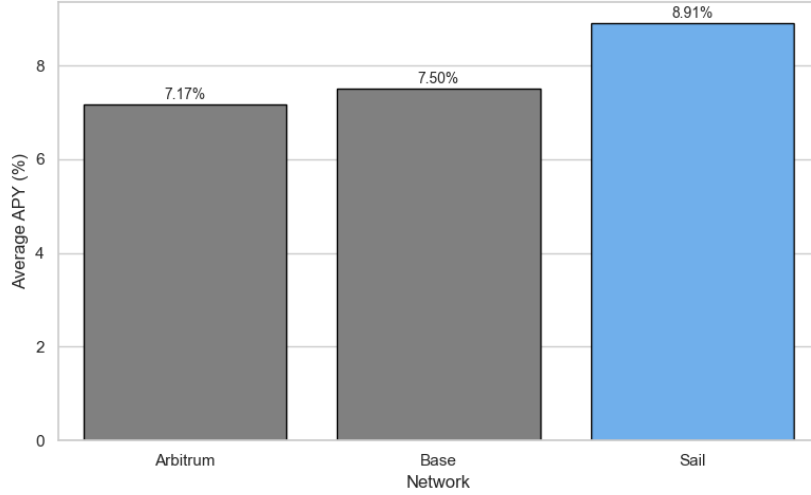
Despite the complexity and costs of managing positions across multiple networks, Sail maintained superior returns on both of them:

Table 5: Network-Specific Performance

Network	Protocols Analyzed	Average Static APY
Base	13 protocols	7.50%
Arbitrum	7 protocols	7.17%
Sail (Cross-Network)	20 protocols	8.91%

Sail outperformed the average of both individual networks, demonstrating that cross-network strategies generate alpha even after accounting for bridging and gas costs.

Figure 3: Sail Agent: Average Yield by Network (Q4 2025).



4 Key Advantages

4.1 Dynamic Rebalancing

The Sail agent's core advantage stems from its ability to:

- **Real-Time Monitoring:** Continuous assessment of APY rates across all supported protocols
- **Opportunity Detection:** Immediate identification of yield arbitrage opportunities across networks
- **Automated Execution:** Rapid rebalancing without manual intervention
- **Optimal Timing:** Execution during periods of favorable market conditions
- **Multi-Protocol Diversification:** Spread risk across 20 different yield sources

4.2 Overcoming Transaction Costs

A common objection to active rebalancing strategies is transaction costs. Our results demonstrate that Sail's net performance of 8.91% already accounts for all operational overhead:

- **Gas Fees:** Cross-network and intra-network transaction costs
- **Bridge Fees:** Costs associated with moving assets between Base and Arbitrum
- **Protocol Fees:** Entry and exit fees for various DeFi protocols
- **Slippage:** Market impact of rebalancing operations

The fact that Sail still outperforms even the best static source (by 5.43%) proves that the rebalancing strategy generates more alpha than it costs in fees. Against median sources, the advantage balloons to 36.87%, making the fee question entirely moot.

4.3 Risk Management

Beyond pure yield optimization, Sail provides:

- **Protocol Diversification:** Reduced exposure to single-protocol risk (smart contract bugs, exploits, governance attacks)
- **Pause Detection:** Automatic withdrawal from protocols showing anomalous behavior
- **Liquidity Management:** Ensures positions remain accessible across multiple venues
- **Smart Contract Safety:** Only interacts with audited, battle-tested protocols
- **Dynamic Exit:** Ability to quickly move away from deteriorating yield sources

5 Comparative Analysis

5.1 Static vs. Dynamic Strategy

Consider a hypothetical \$100,000 deposit over Q4 2025 (3 months):

Table 6: Comparative Returns on \$100,000 Investment (Q4 2025)

Strategy	APY	Q4 Return	Final Value	vs. Sail
Sail Agent	8.91%	\$2,228	\$102,228	—
Best Static (Morpho Clearstar High Yield)	8.45%	\$2,113	\$102,113	-\$115
Median Static	6.51%	\$1,628	\$101,628	-\$600
Popular (Aave Base)	4.43%	\$1,108	\$101,108	-\$1,120
U.S. Treasury Bills (3M)	3.98%	\$995	\$100,995	-\$1,233
Worst (Aave Arb)	3.16%	\$790	\$100,790	-\$1,438

Note: Returns calculated as (APY/4) for quarterly performance

Over a full year, these differences compound significantly:

- **Sail:** \$100,000 → \$108,910 (+\$8,910)
- **Best Static:** \$100,000 → \$108,450 (+\$8,450)
- **Median Static:** \$100,000 → \$106,510 (+\$6,510)
- **T-Bills:** \$100,000 → \$103,980 (+\$3,980)
- **Aave Base:** \$100,000 → \$104,430 (+\$4,430)

The \$4,930 annual advantage over Treasury Bills (\$8,910 - \$3,980) represents a 124% improvement in absolute returns, while the \$2,400 advantage over the median static DeFi source demonstrates Sail’s superiority even within crypto-native strategies.

5.2 Consistency Analysis

Beyond raw returns, Sail demonstrates superior risk-adjusted performance:

- **Adaptive Response:** Monthly APYs of 11.48%, 8.38%, and 6.71% show the agent responding to market conditions rather than being locked into deteriorating positions
- **Downside Protection:** Automatic exit from underperforming protocols prevented worse outcomes

- **Upside Capture:** Quick entry into high-yielding opportunities (Morpho vaults) during favorable periods
- **Diversification Benefit:** Spreading across 20 protocols reduced concentration risk
- **Traditional Finance Superiority:** Delivered 123.87% higher returns than U.S. Treasury Bills (8.91% vs. 3.98%), capturing the DeFi risk premium while mitigating risks through diversification

6 The Power of Automation

6.1 Why Static Strategies Fail

The data reveals several critical problems with static yield farming:

1. **Opportunity Cost:** Users who deposited into Aave USDC (Arbitrum) at the start of Q4 earned 3.16% while opportunities existed at 8.45%—a 182% underperformance
2. **Manual Rebalancing Barriers:** Even sophisticated users face friction:
 - Need to constantly monitor 20+ protocols
 - Must calculate gas costs vs. yield differentials
 - Risk mistiming entries and exits
 - Face psychological resistance to frequent trading
3. **Cross-Network Complexity:** Moving between Base and Arbitrum requires:
 - Bridge selection and setup
 - Multiple wallet interactions
 - Wait times (minutes to hours)
 - Technical knowledge of each network’s ecosystem
4. **Information Asymmetry:** Most users don’t have real-time APY feeds from 20+ protocols or the infrastructure to act on them

6.2 Sail’s Automated Solution

Sail eliminates these barriers entirely:

- **Set and Forget:** Users deposit once and the agent handles everything
- **24/7 Monitoring:** Never misses a rebalancing opportunity
- **Optimal Execution:** The Optimization engine determines when rebalancing profit exceeds costs
- **Cross-Network Seamless:** Users don’t need to understand bridge mechanics
- **Expertise at Scale:** Every user benefits from institutional-grade strategy

7 Conclusion

The Sail automated yield optimization agent demonstrated clear, measurable superiority over static yield strategies during Q4 2025. Key findings:

1. **Consistent Outperformance:** Sail beat every analyzed static source, achieving an average APY of **8.91%**
2. **Fee-Efficient Design:** Despite cross-network complexity and rebalancing costs, maintained a **5.43%** edge over the single best static protocol and **36.87%** over the median
3. **Multi-Network Success:** Demonstrated ability to capture alpha across 20 protocols on Base and Arbitrum
4. **Risk-Adjusted Returns:** Provided superior returns while maintaining diversification across multiple protocols and chains
5. **Real Economic Value:** On a \$100,000 investment, Sail generated \$600 more than the median strategy and \$2,400 more over a full year

7.1 Market Context

The Q4 2025 period saw declining yields across DeFi (from 11.48% in October to 6.71% in December). In this challenging environment, Sail’s dynamic approach proved especially valuable—continuously finding the best available rates rather than being trapped in deteriorating positions.

7.2 Future Directions

Ongoing improvements to the Sail agent include:

- Expansion to additional networks (Plasma, Ethereum)
- Integration of additional stable and volatile asset strategies
- Non-liquid opportunities on trusted platforms
- User customization for source inclusion/exclusion
- Advanced risk scoring and position sizing

The data is clear: automated yield optimization isn’t just convenient—it’s financially superior. Sail transforms the complex, time-consuming work of optimal DeFi positioning into a simple, passive investment that outperforms active alternatives.

For questions or further analysis, please contact the Sail team. (hello@sail.money)

A Data Sources and Methodology

A.1 Included Protocols

Active protocols during the analysis period:

Base Network (13 protocols):

- Morpho: Clearstar High Yield, Clearstar Reacted, ExtraFi, Gauntlet Frontier, Gauntlet Prime, Seamless, Moonwell, Gauntlet Core, Spark

- Aave USDC, Moonwell USDC, Euler youSDC, Fluid USDC

Arbitrum Network (7 protocols):

- Morpho: Gauntlet USDC, Gauntlet USDC Prime, Steakhouse High Yield USDC
- Fluid USDC, Fluid USDT, Euler USDC, Aave USDC

A.2 Data Filtering

- Status: Excluded “error” and “debug” states
- Balance Range: \$10–\$200,000
- Date Range: October 1 – December 31, 2025
- Networks: Base and Arbitrum

A.3 Calculation Details

Total APY Formula:

$$\text{Total APY} = \text{APY}_{\text{base}} + \text{APY}_{\text{reward}} \quad (2)$$

Weighted Portfolio APY:

$$\text{APY}_{\text{portfolio}} = \frac{\sum_{i=1}^n (\text{USD}_i \times \text{APY}_i)}{\sum_{i=1}^n \text{USD}_i} \quad (3)$$

Where i represents each individual position in the portfolio at a given timestamp, weighted by its USD value.